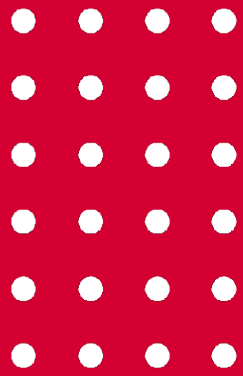


Student Perspective: Using AI to Create Individualized, Accessible Reading Materials

Sarah Chew and Bridget Brown
with Marissa Bloodgood and Eric
Boulahanis





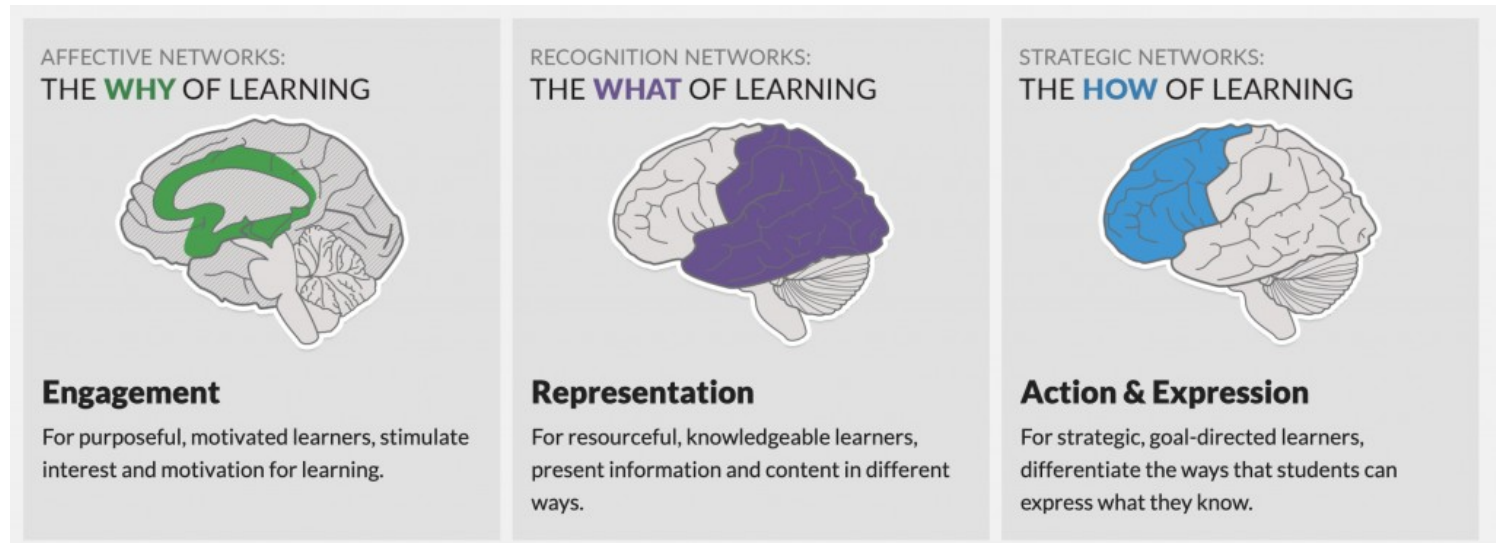
Challenge 2022-2023

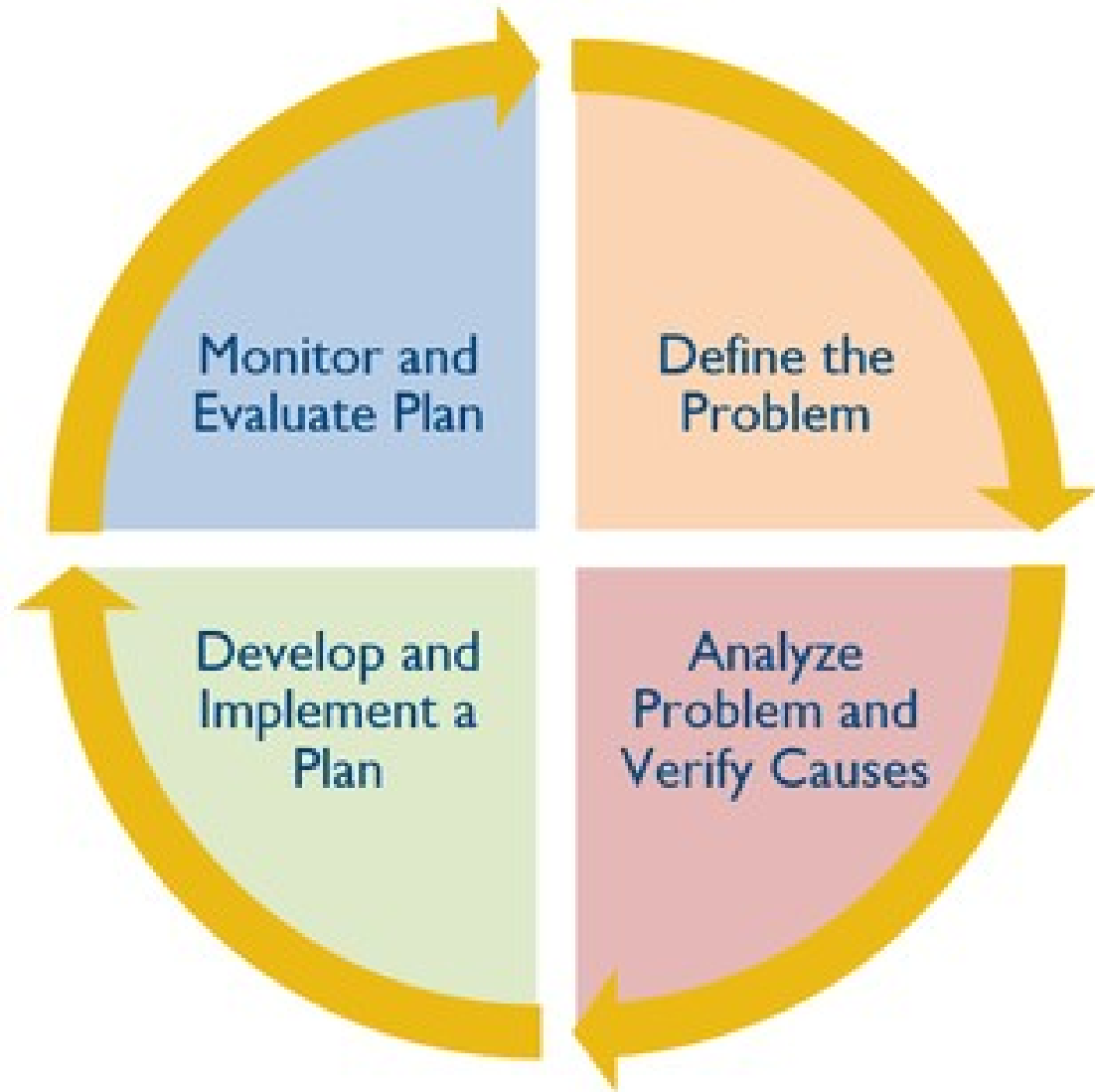
How to provide access to integrated classes when modifications and auditing are not an option

Original Solution

Manually create “plain language”

- Time-consuming
- Not individualized nor consistently “plain language”
- Doesn’t prepare students for post-college (can they expect employers to do this?)
- UDL is multiple means of engagement, representation, & action and expression but this only covers representation





Revised Solution

Have staff use AI to create plain language

Only followed problem solving process, not JHU AT Cycle



Monitor & Evaluate Plan

Spring 2023

New priority problem to focus on: **how does this transfer to students' lives outside of school?**

Begin teaching students AI-assisted text simplification process

- Too many steps
- Telling AI “plain language” or “simplify” doesn’t get close enough but using grade level terms doesn’t feel empowering
- If students don’t know their reading level, what do they enter?
- Not co-created with students so missing their perspectives on quality of process

Using AI to Simplify Writing

Plain language and Easy Read are writing styles that are designed to increase reader comprehension. Plain Language is generally at a 6th-8th grade reading level while Easy Read is 5th grade or lower. 3rd grade is a pivotal year in students' reading development, so it is a good place to start when creating materials accessible to a variety of people with intellectual disabilities.

IDENTIFY A DIGITAL TEXT YOU WANT TO CONVERT AND COPY THE TEXT

- ☐ If you are writing an original text, write it in your typical writing voice, then use a tool like ChatGPT to convert it to plain language.
- ☐ The Mac keyboard shortcut to copy is Command + C.
- ☐ The PC keyboard shortcut to copy is Control + C.

IDENTIFY ANY WORDS/PHRASES YOU WANT THE AI TO KEEP OR CHANGE

- ☐ This helps you quickly check if what the AI gave you matches your goals
- ☐ Words to keep might be proper nouns or other necessary vocabulary.

GO TO YOUR CHOSEN AI, WRITE “REWRITE THIS AT A 3RD-GRADE LEVEL,” PASTE THE ORIGINAL TEXT, AND PRESS ENTER

- ☐ Being specific to say 3rd grade gives the AI a clear idea of what you want while using the prompt “simplify” may result in an oversimplified output or a still too complex output.
- ☐ The Mac keyboard shortcut to paste is Command + V.
- ☐ The PC keyboard shortcut to paste is Control + V.

ADJUST UNTIL YOU GET THE TEXT YOU WANT

- ☐ You may give the AI additional prompts like “rewrite at a 2nd-grade level” or “rewrite but keep X words”, or manually make changes.

OPTIONAL: ENTER THE TEXT INTO HEMINGWAY (FREE)

- ☐ Pasting the new text into hemingwayapp.com gives a grade-level readability score and suggestions on how to make the writing even clearer.

FORMAT TO SHARE

- ☐ Add final touches! You can go an extra step and edit the text so that there is only one idea per line. The Autistic Self-Advocacy Network has a resource called “One Idea Per Line” to support manual formatting.

Have any questions? Reach out to Sarah Chew at schew4@uic.edu



Solution entering SY 2023-2024

Determine each student’s independent reading comprehension* level

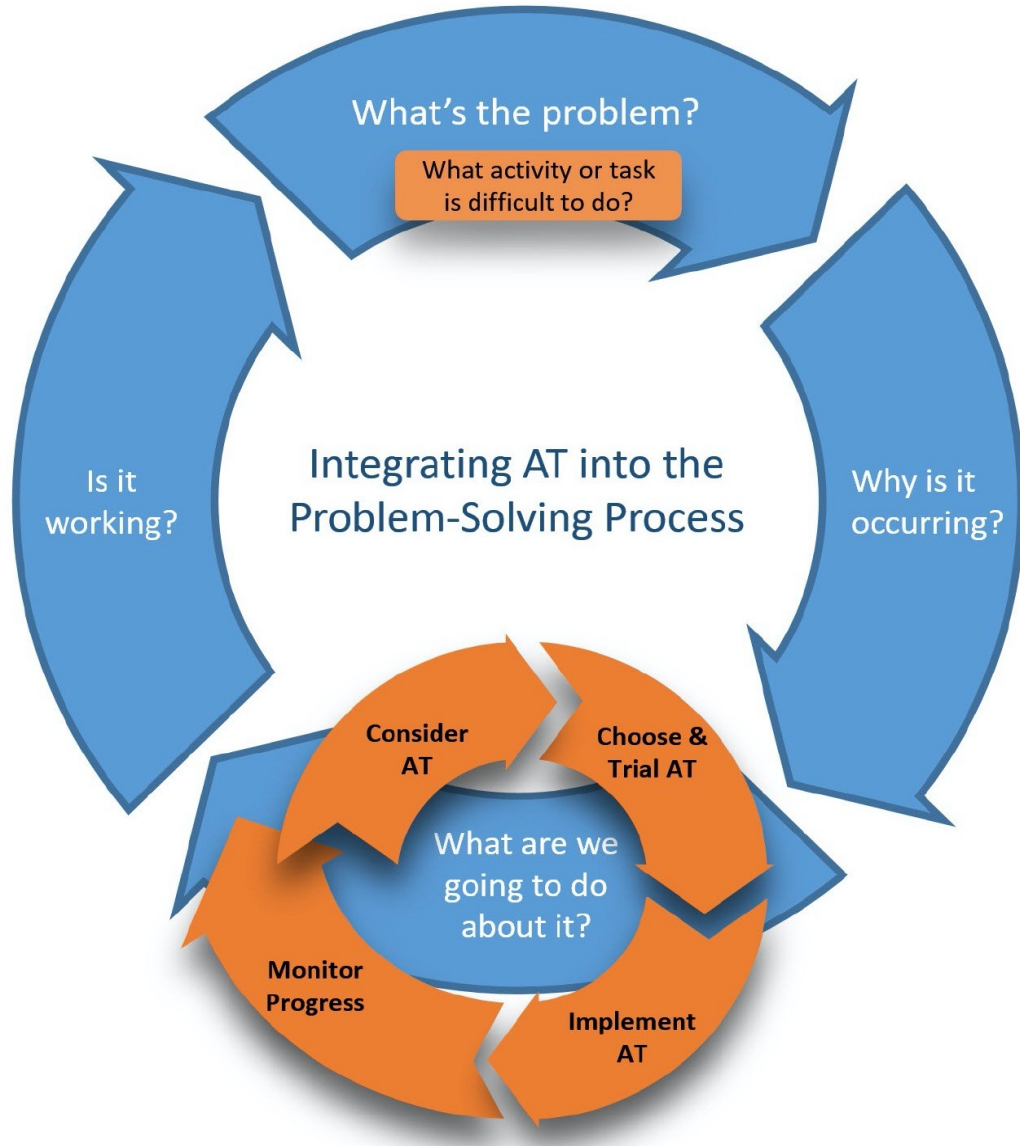
Teach them how to use generative AI to convert reading materials to their reading level in a 1:1 setting to individualize the process as well

Common Core Aligned Text Correlation Chart

Grade Level	Lexile Level	F & P	STEP	QRI
Primer	BR		Pre-Primer	Pre-Primer
Kinder	BR	A, B, C, D, E	1, 2, 3	Primer, 1 st
1	190L to 530L	E, F, G, H, I, J, K	4, 5, 6	1 st
2	420L to 650L	J, K, L, M, N, O	7, 8, 9	2 nd , 3 rd
3	520L to 820L	N, O, P, Q, R	10, 11, 12	4 th
4	740L to 940L	Q, R, S, T		5 th , 6 th
5	830L to 1010L	U, V, W		6 th
6	925L to 1070L	X, Y, Z		Upper Middle School
7	970L to 1120L			
8	1010L to 1185L			
9	1050L to 1260L			High School
10	1080L to 1335L			
11 and 12	1185L to 1385L			

Independent reading level (90% or higher accuracy)
Instructional reading level (70-89% accuracy)
Frustration reading level (Below 70% accuracy)





Use PAR to determine if AI actually helps

Protocol for Accommodations in Reading

1. Find independent reading level
2. Test HS passage with human reader (remove decoding demands, keeps comprehension demands)
3. Test HS passage with Chat GPT text simplification (keep decoding demands, reduce comprehension demands)

Finding Student Reading Comprehension Levels

Grade Level	Student A	Student B	Student C
High School			
Middle School			
6			
5			
4	50% accuracy Narrative text		37.5% accuracy Expository text
3	50% accuracy Expository text 75% accuracy Narrative Text	87.5% accuracy Expository text	25% accuracy Expository text
2	50% accuracy Expository text 62.5% accuracy Narrative text	100% accuracy Expository text	87.5% accuracy Expository text
1	100% accuracy Expository text		100% accuracy Expository text

3 other students tested

- 1 Pre-reader (learning to decode)
- 1 1st grade level
- 1 2nd grade level

This tells us neither “plain language” nor “easy read” are a fit for our students.

Plain language = ~6th-7th grade

Easy read = ~3rd-5th grade

Does AI Help? Student A's data from PAR

Grade Level	Oral Read by Student	Teacher Read with Student Following Along with a Copy	Oral Read by Student with ChatGPT 3.5 Text Simplification
High School		20% Accuracy WWI Part One	70% Accuracy WWI Part Two
Middle School			
6			
5			
4	50% accuracy Narrative text		
3	50% accuracy Expository text 75% accuracy Narrative Text		
2	50% accuracy Expository text 62.5% accuracy Narrative text		
1	100% accuracy Expository text		

Yes!

Moved student to 70% accuracy from 20% accuracy in one sitting.

Need to reduce or remove some other task demand to get student to independent reading comprehension

Does AI Help? Student B's data from PAR

Grade Level	Oral Read by Student	Teacher Read with Student Following Along with a Copy	Oral Read by Student with ChatGPT 3.5 Text Simplification
High School		30% Accuracy WWI Part One	90% Accuracy WWI Part Two
Middle School			
6			
5			
4			
3	87.5% accuracy Expository text		
2	100% accuracy Expository text		
1			

Yes!

Moved student to 90% accuracy from 30% accuracy in one sitting

Monitor to see if this trend continues

Does AI Help? Student C's data from PAR

Grade Level	Oral Read by Student	Teacher Read with Student Following Along with a Copy	Oral Read by Student with ChatGPT 3.5 Text Simplification
High School		0% Accuracy WWI Part One	20% Accuracy WWI Part Two
Middle School			
6			
5			
4	37.5% accuracy Expository text		
3	25% accuracy Expository text		
2	87.5% accuracy Expository text		
1	100% accuracy Expository text		

Yes—but not enough

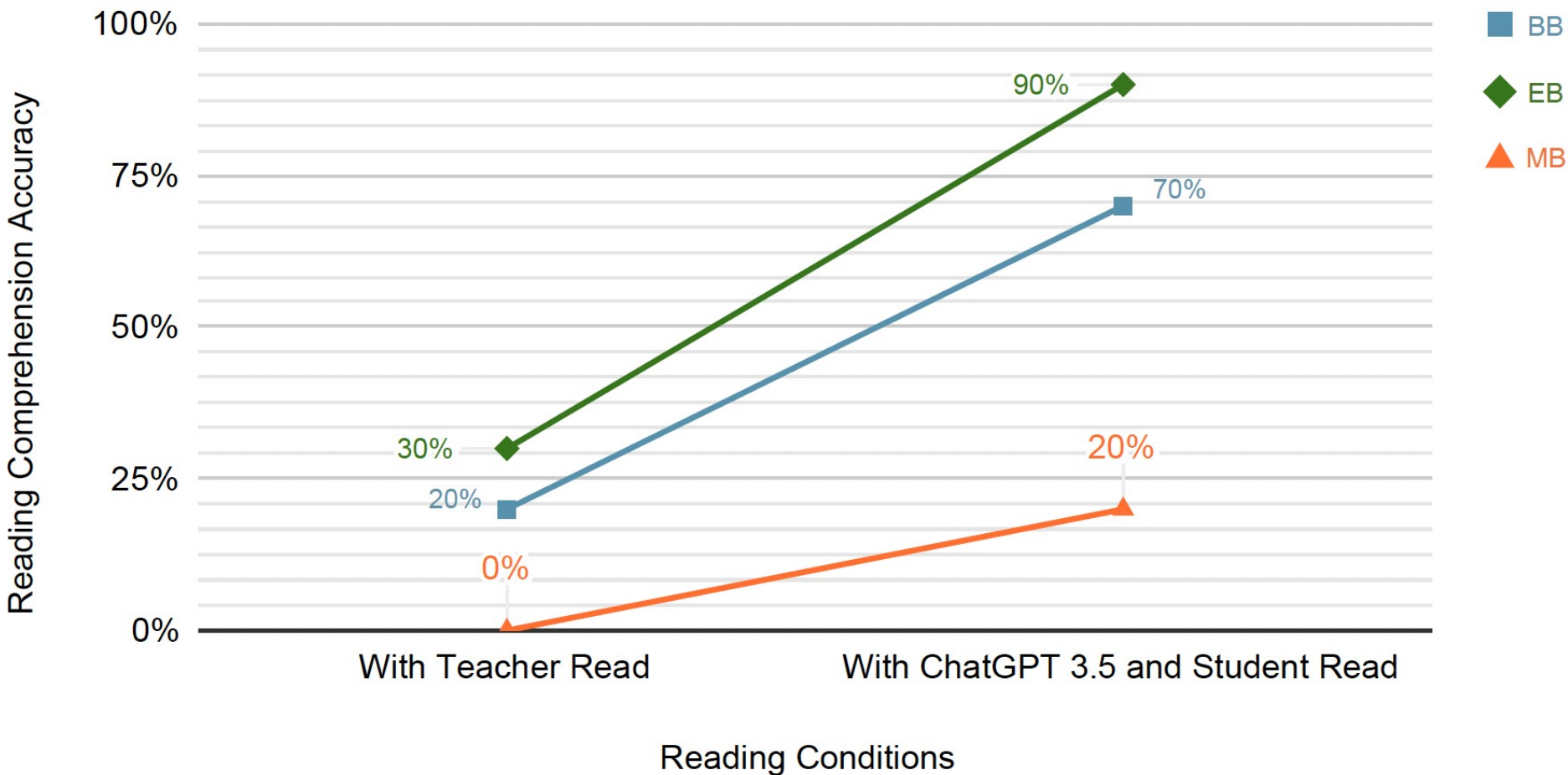
Moved student to 20% accuracy from 0% accuracy in one sitting

Continue evaluation to see what task demands can be reduced or removed to support this student

Confirmed: Individualized AI Text-Simplification Helps*

The Effect of Text Simplification on Reading Accuracy

Without Simplification or Technology vs. With Simplification and Technology



Customize ChatGPT

Custom Instructions ⓘ

What would you like ChatGPT to know about you to provide better responses?

I have an intellectual disability and use ChatGPT to help me understand difficult words or reading passages



107/1500

How would you like ChatGPT to respond?

Write all responses at a Fountas and Pinnell level J, or 2nd-grade, reading comprehension level

Enable for new chats ☒

Cancel

Save

Important Notes

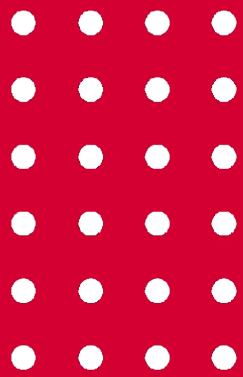
Ask if a student prefers dark/light appearance

Use Settings > Personalization > Customize

Chat GPT has built-in Text-to-Speech

Add Alternate Format and Open Notes to Letter of Accommodation, if possible

Practice, practice, practice



Challenge 2024

Reading levels are phrased in terms of grade-level equivalency. Telling an adult they read at an elementary grade level can be disempowering



Solution

Gain perspective from people with a variety of college experience who will be using the tool and refine the process

(Videos)



Bridget

Co-Op student graduating Fall 2024—has been pursuing college for 20+ years. From a family involved in academia



Eric

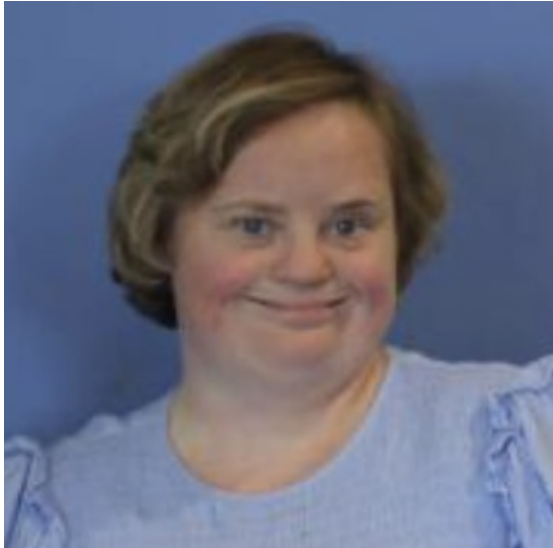
Would have gone to college but didn't know it was an option; has participated in leadership programs; works ~2 hours/week



Marissa

Graduated from 6 month non-inclusive program; starting a 2 year commuter program in the Fall with the goal of going to a 2 year residential program

What are the benefits of using AI to change reading materials to be closer to a student's independent reading level?



Bridget

Success on the exam and understand the material



Eric

It's fast—you just paste it in and it's on your level

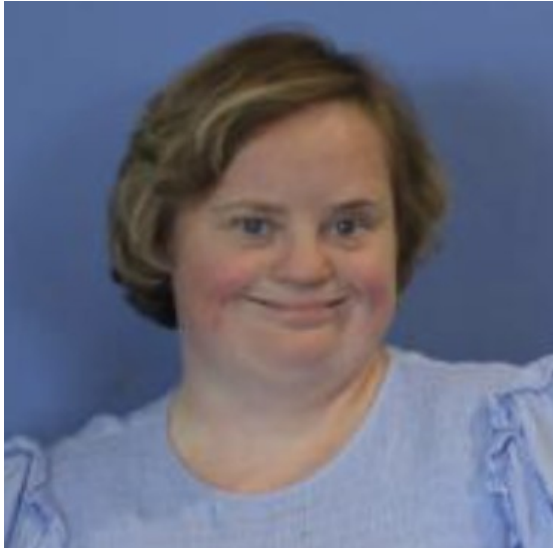


Marissa

It helps me understand—the passages can still be hard but it helps me understand

What do you think?

Some people are concerned using AI is cheating. How does allowing students to use AI support inclusive post-secondary education?



Bridget

This isn't cheating because I'm still responsible for knowing the information.



Eric

Kinda cheating but not really—AI is like an aide but it's on a computer instead of having a person

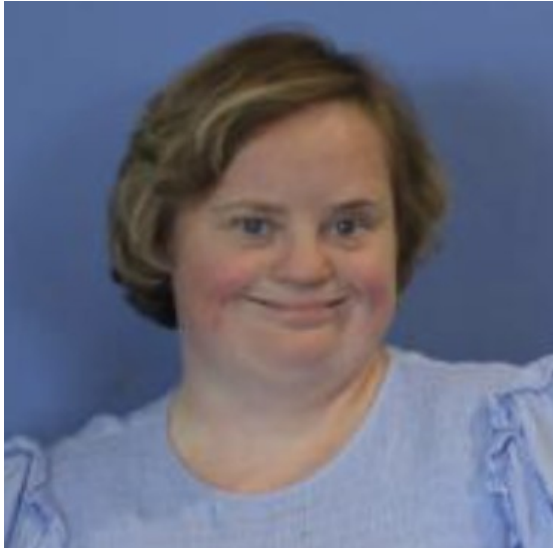


Marissa

I don't think it's cheating—you still have to answer the same questions. This lets me share with discussion partners

What do you think?

What are the benefits of teaching students to use AI independently?



Bridget

It transfers to your job; what the teacher makes might still be too hard; you can better match your strengths



Eric

Flexibility & individualization: things on my reading level can still be hard if it's told a difficult way



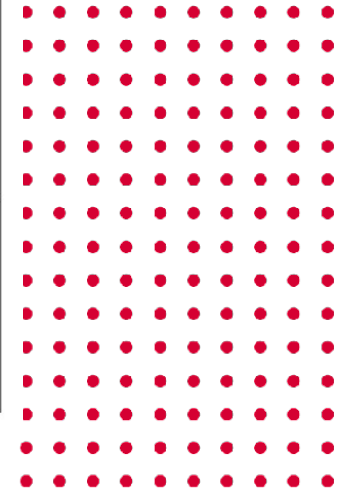
Marissa

The teacher version can still be too hard; I can best match me and be better able to do assignments.

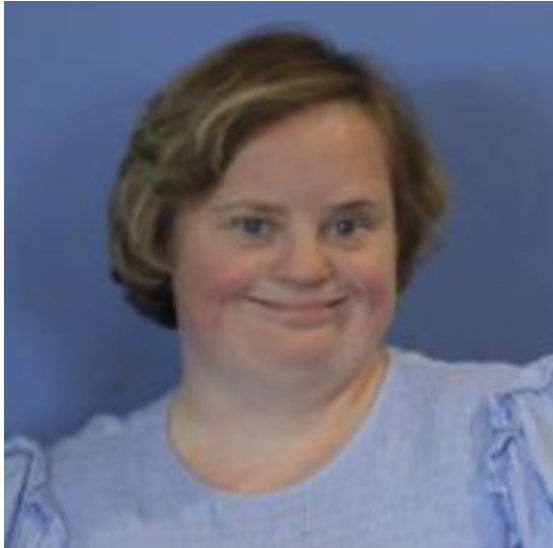
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When setting the AI, should a prompt that uses letters be used instead of numbers? Why or why not?



Bridget

Letters: It could be harmful to someone's self-confidence to tell them they read at a 1st grade level or 2nd grade level.



Eric

Letters: If someone told me I read at a first grade level I would say "I don't care"—but letters are easier

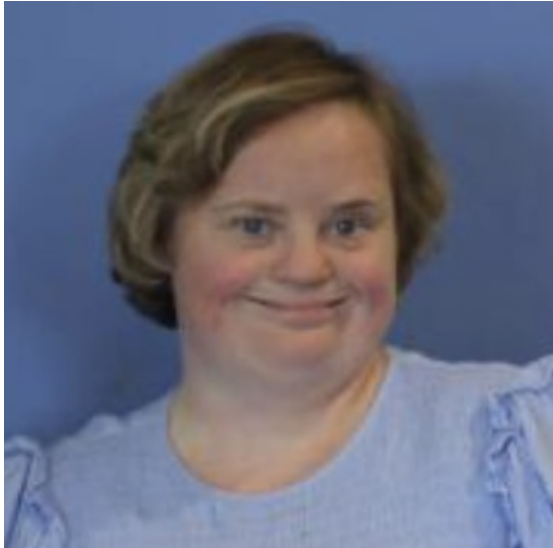


Marissa

The letters are better than the grade levels because they can find what they need and it won't make them upset.

What do you think?

Should the student need to remember their own independent reading level or should the AI setting be used so they can just say "rewrite this for me?"



Bridget

"Rewrite this for me" is easier than remembering "Fountas and Pinnell"



Eric

Both—it's helpful to remember your reading level if you want to read something not on the AI



Marissa

Set the AI settings. They can always go check the settings if they want to find their reading level

What do you think?



Questions?

Thank you!